GradeSense - Model Answer Paper

Q1 — Science

Model Answer — 5 marks

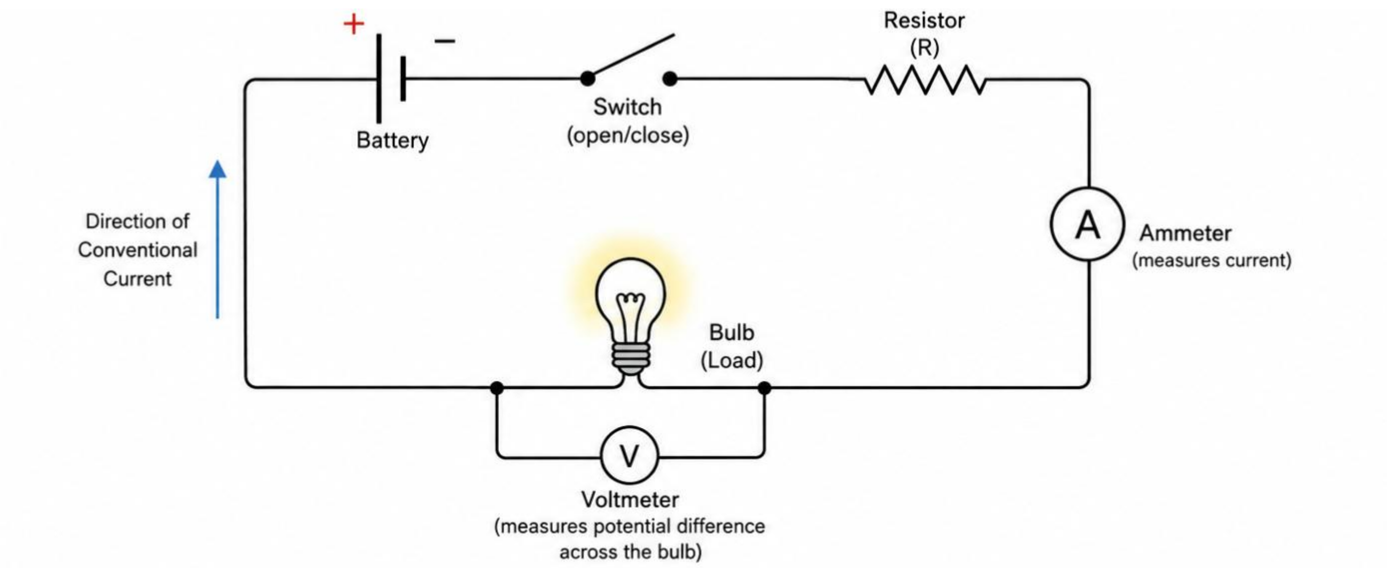
A simple electric circuit provides a closed path through which electric current can flow. The battery provides the potential difference that drives current through the circuit. The switch is used to open or close the circuit. When the switch is closed, there is a complete conducting path and current can flow through the bulb and resistor. When the switch is open, the path is broken and current does not flow.

The bulb, resistor, battery, switch and ammeter should be connected **in series** in the main circuit. The ammeter must be connected in series because it measures the current flowing through the circuit. A voltmeter, however, should be connected **in parallel across the bulb**, because it measures the potential difference between the two ends of the bulb.

A properly labelled diagram would therefore show the battery connected to the switch, resistor, bulb and ammeter in a single closed loop, with the voltmeter connected across the bulb. The conventional current direction should be shown from the positive terminal of the battery through the external circuit towards the negative terminal.

The amount of current flowing through the circuit depends on both the potential difference and the resistance. According to Ohm's law, **V = IR**. If the voltage of the battery remains constant and the resistance is increased, the current flowing through the circuit decreases. Conversely, reducing the resistance allows more current to flow.

What the diagram should contain



A full-credit diagram should visually represent something equivalent to:

- Battery
- Switch
- Resistor
- Bulb
- Ammeter **in series**
- Voltmeter **in parallel across the bulb**
- Connecting wires forming a complete circuit
- Appropriate component labels

- Conventional current direction

Marking rubric

Criterion	Marks
Correctly represents the main circuit with battery, switch, bulb and resistor connected in a closed series path	1
Correct placement of ammeter in series and voltmeter in parallel across the bulb	1
Correct explanation of current flow and the function of the main components	1
Correctly explains the relationship between resistance and current, including the relevant principle/Ohm's law	1
Clear, logically structured explanation with appropriate labels and current direction in the diagram	1
Total	**5**

Important grading guidance

The student **does not need to reproduce the model answer word-for-word**.

For example:

"If resistance increases, less current will flow because the voltage remains constant."

should receive credit even without explicitly writing Ohm's law.

Likewise, a student may draw the circuit using slightly different physical layouts while still receiving full marks if the **electrical relationships are correct**.

However, if the student places the voltmeter in series with the bulb, that is a substantive error and should affect the relevant rubric mark.

Q2 — English

Model Answer — 5 marks

Technology has significantly changed the way students learn because information that was once difficult to access is now available within seconds. Students can use educational websites, digital libraries, videos and interactive tools to understand concepts that they may not have understood in a classroom. Technology can therefore make learning more flexible and allow students to explore topics according to their own interests and pace. For example, a student struggling with a difficult scientific concept can watch several different explanations until they find one that they understand.

However, having easy access to information can also create a problem. Students may become dependent on searching for an answer instead of trying to solve a problem themselves. If a student immediately looks up the solution to every difficult question, they may complete their work but fail to develop their own reasoning and problem-solving abilities. There is also a risk that students may accept inaccurate information simply because it appears online.

In my opinion, technology does not automatically make students better or worse learners. Its effect depends on how it is used. Technology is most useful when students use it to understand concepts, explore different ideas and check their own work rather than simply copying answers. Therefore, technology should be treated as a tool that supports thinking, not as a replacement for thinking.

Marking rubric

Criterion	Marks
Presents a clear position on whether/how technology affects learning	1
Provides relevant and logically developed arguments supporting the position	1
Recognises and meaningfully addresses an opposing viewpoint or limitation	1
Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims	1
Provides a coherent conclusion that follows from the discussion, with clear overall communication	1
Total	**5**

Important grading guidance

This question is intentionally **open-ended**.

A student does **not** have to reach the same conclusion as the model answer.

For example, a student could argue:

"Technology makes students worse learners because they become dependent on instant answers."

That can still receive **5/5** if the student develops the argument properly, gives examples, considers the opposing viewpoint and reaches a logical conclusion.

Similarly, a student who strongly argues that technology improves education should not lose marks simply because their position differs from the model answer.

This is an important test of whether the grading system evaluates **quality of reasoning rather than similarity to the model answer**.

Q3 — Economics

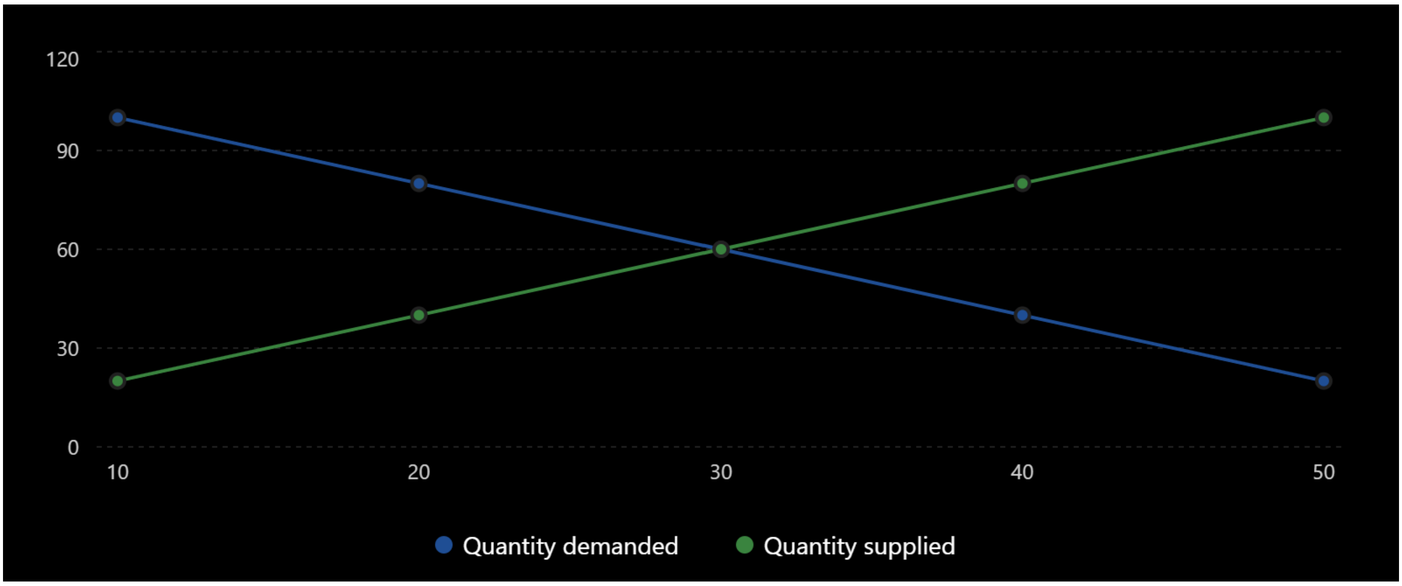
Model Answer — 5 marks

The demand and supply data can be represented on a graph with **quantity on the horizontal axis** and **price on the vertical axis**. The demand curve should slope downward from left to right because consumers generally demand a greater quantity when the price is lower. The supply curve should slope upward from left to right because producers are generally willing to supply a greater quantity when the price is higher.

The two curves intersect at a price of **₹30** and a quantity of **60 units**. This point represents the market equilibrium because at this price the quantity demanded is equal to the quantity supplied.

Demand and supply schedule

The given market data show quantity demanded falling as price rises and quantity supplied increasing.



The equilibrium occurs where the two schedules meet: ₹30 and 60 units.

If the market price is **below the equilibrium price**, quantity demanded is greater than quantity supplied. This creates a shortage because consumers want to buy more than producers are willing to sell. The shortage creates upward pressure on price.

If the market price is **above the equilibrium price**, quantity supplied is greater than quantity demanded. This creates a surplus because producers have more goods available than consumers are willing to purchase. The surplus creates downward pressure on price.

If the cost of producing the product increases, production becomes less profitable at each existing price. Producers will therefore be willing to supply less at each price. The supply curve shifts **to the left/upward**, from the original supply curve to a new supply curve.

The new equilibrium would generally occur at a **higher price and lower quantity**, assuming demand remains unchanged.

Marking rubric

Criterion	Marks
Correctly plots and labels the demand and supply curves, with appropriate axes and direction	1
Correctly identifies the equilibrium at ₹30 and 60 units and explains why it is equilibrium	1
Correctly explains shortage below equilibrium and surplus above equilibrium	1
Correctly explains that increased production costs shift the supply curve left/upward	1
Correctly explains the resulting tendency toward a higher equilibrium price and lower equilibrium quantity, with the change represented appropriately on the graph	1
Total	**5**